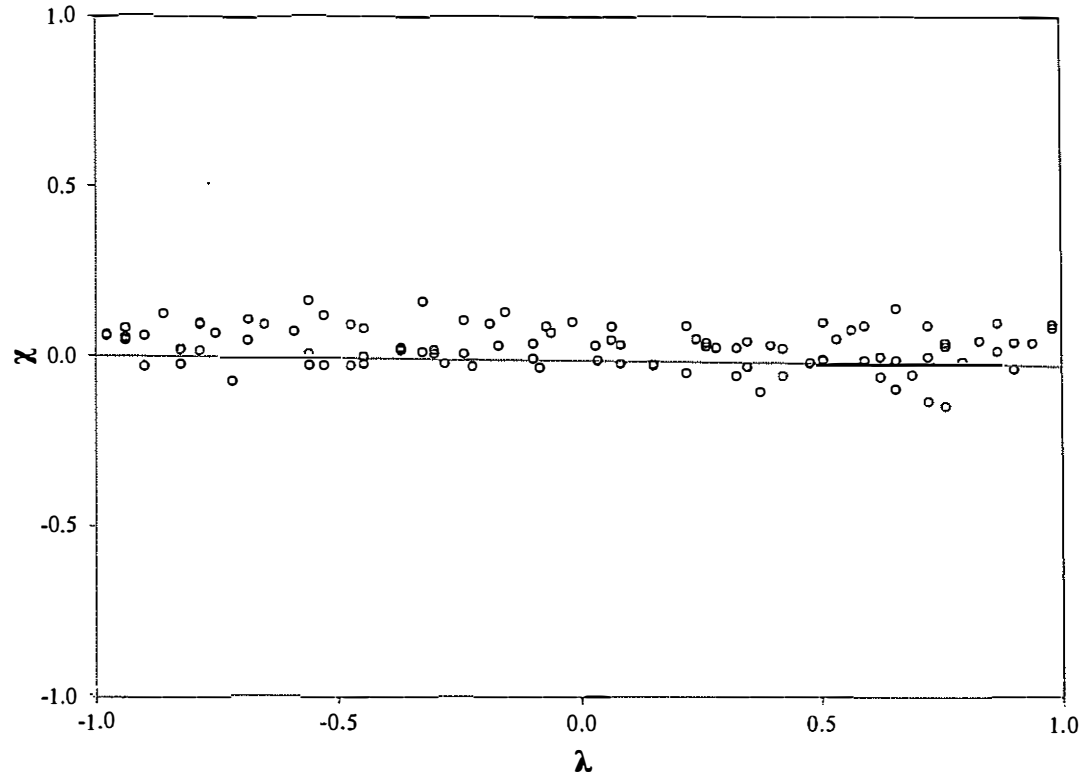


EXAM 7 – SPRING 2012

16. (1.5 points)

A χ -plot of indemnity amounts and ALAE amounts for auto physical damage claims is shown below.



a. (1 point)

Explain what the chi (χ) and lambda (λ) variables measure.

b. (0.5 point)

Based on the χ -plot, draw a conclusion regarding the auto physical damage book of business.

CONTINUED ON NEXT PAGE

Question 16 Sample Answer

Solution 1

a)

$$\chi_i = \frac{H_i - F_i G_i}{\sqrt{F_i(1-F_i)G_i(1-G_i)}} \quad \begin{array}{l} \text{measures how far the joint distribution} \\ \text{is from the independent distribution} \end{array}$$
$$\lambda = 4 \times \text{sign}(\tilde{F}_i \tilde{G}_i) \times \max(\tilde{F}_i^2 \tilde{G}_i^2) \quad \lambda_i =$$

where $\tilde{F}_i = F_i - \frac{1}{2}$ and $\tilde{G}_i = G_i - \frac{1}{2}$

F_i and G_i use the marginal distribution and H_i is the joint distribution

λ measures where in the distribution with H indicating the top left or bottom right and 1 indicating the top right or bottom left, and 0 indicates near the mid point.

Solution 2

a) χ measures independence, since the numerator of the χ calculation takes joint distribution minus the product of the marginals. This would be 0 for independence.

λ measures distance to the center of the distribution.

Solution 3

a) χ measures the difference between the value of the empirical copula at a specific point (H) and the null hypothesis of independence of the two distributions ($F \cdot G$).

$$\chi = \frac{H - FG}{\sqrt{F(1-F)G(1-G)}} \quad \begin{array}{l} \rightarrow \text{difference described above} \\ \rightarrow \text{scaling factor to keep between } -1 \text{ and } 1 \end{array}$$

λ is the measure of the signed distance between the point and the point in the bivariate distribution with the median coordinates, where F and $G = .5$

Solution 1

b) If the bulk of the observations are near the $\chi=0$ line then the 2 series are independent. Based on this chart the auto PD book is independent for the losses and ALAE.

Solution 2

- b) There appear to be more positive X than negative ones. If they were randomly scattered around 0 you could say that indemnity and ALAE were independent, but with more positives than negatives it looks like the empirical relationship shows a little more dependence than none. The dependence does not appear to vary though, along the distribution. It appears to be steady.

Solution 3

- b) The data points are uniformly random around the $x=0$ line, this implies they are independent. If there was a pattern (like an arc), then we would draw a different conclusion.

Examiner Comment

A few candidates defined lambda as the distance from the mean of the distribution rather than the median. Also, some candidates failed to illustrate the signed nature of lambda.

A few candidates defined chi as a residual and while mathematically correct, this definition does not provide any insights into how chi measures dependency.

A few candidates did not adequately explain their conclusions from the graph.